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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

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1. Team & Project Overview

Team Name	Made in UTM
Selected Domain	Domain 1: The Aiversecity Future-Ready Workforce
Selected SDG Track (1–17)	SDG 4 Quality Education (main track, targets 4.3 and 4.4), with SDG 8 Decent Work and Economic Growth (targets 8.2 and 8.5)
Project / Solution Name	Laluan (Malay for "path"): a career GPS for the AI era. Live: https://laluan-psi.vercel.app
Name (Leader)	Tan Hong Sheng
Email & Phone (Leader)	tanhongsheng050204@gmail.com +60 11-1691 8805
University Affiliation(s)	Universiti Teknologi Malaysia (UTM)
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI is changing which jobs exist faster than training programmes and skill frameworks can keep up. In Malaysia, 56% of professionals already use AI tools at work, yet only 26% have had formal training (Hays Asia Salary Guide 2025, cited in the Domain 1 brief), and MOSTI expects more than 600,000 workers to need reskilling within three to five years. Globally, the World Economic Forum expects a structural churn of 22% of formal jobs by 2030 (170 million created, 92 million displaced), with clerical roles shrinking fastest: data entry clerks by about 26% and administrative assistants by about 20% (Future of Jobs Report 2025, Figures 2.1 and 2.3).

Those affected first have the least room to experiment: clerks, customer-service staff and other routine-task workers, graduates choosing a first job, and HR teams deciding whom to reskill. Today they rely on static career lists that date quickly, one-off assessments, or generic chatbots whose advice cannot be traced to any source.

This matters socially and economically. A worker who reskills too late loses income and confidence; a graduate who trains for a shrinking role wastes time and money; a company that guesses wrong spends its training budget on the wrong skills. What is missing is guidance that is **personal, current, explainable and honest about uncertainty**: it should show which skills carry over, which roles are likely to grow under different AI-adoption futures, and an affordable plan to get there, without making the decision for the person.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Aisyah, a data-entry clerk, learns her team adopted an AI invoice reader. Laluan takes her from “who I am today” to “where I can be in the AI era” in minutes.

1. **Skills in.** CV upload, questionnaire or skill list. Names, IC and phone numbers are removed before any AI reads the text; the file is never stored. She confirms the skills.

2. **Forecasts, not lists.** A re-runnable pipeline collects three years of job-ad mentions, AI-research papers (arXiv) and Malaysian employment (OpenDOSM). Each series becomes a damped trend to 2030 with an

80% range, flagged emerging when growth speeds up. A last-year backtest is shown in the app. Every refresh is dated and **only goes live after a person approves it**.

3. **Three futures.** Slow, baseline and accelerated AI adoption change the forecasts and her ranked roles: 0.6 × skill fit + 0.4 × outlook. “Why this rank?”, a step-by-step forecast x-ray and **tipping points** (the adoption speed where her job turns declining) explain every number.

4. **A plan she can start this week.** Ordered courses for missing skills, with hours, RM price and HRD Corp status from a curated catalogue; AI summaries are checked against it, so nothing is invented.

5. **SDG links with reasons** for every role and skill, and an **SDG lens** showing what choosing for a goal costs in hours and fit; an **AI coach** that only suggests, **automatic re-planning**, a public **fairness check** and an **HR view**.

Hardest risk: reliable, updatable trend data. Fallbacks: noisy trends shrink towards no change, ranges stay visible, arXiv falls back to OpenAlex, and the last approved snapshot stays live.

Workshops used: **GenAI APIs** (Gemini; Qwen2.5 and Whisper on RunPod; offline fallback), **database architecture** (Supabase Postgres) and **cloud deployment** (Vercel, Docker on Render, GitHub Actions).

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: React 18 + TypeScript + Vite; responsive; English, Bahasa Melayu and Chinese; lite mode for slow connections.

Backend API: Python 3.12, FastAPI and Pydantic, packaged as a Docker container.

AI: Google Gemini API (gemini-3.8-flash for the coach, proof-check questions, interview feedback and summaries; gemini-embedding-001 for unfamiliar skill phrases). Open models on RunPod Serverless: Qwen2.5-7B-Instruct reads the scrubbed CV first, so Gemini only sees confirmed skills; faster-whisper for voice input; our own OCR for scanned CVs. A rule-based offline mode runs when no key is set.

Database: Supabase Postgres with row-level security (profiles, progress, Future Cards, trend snapshots, review decisions) and Supabase Auth for optional Google sign-in; versioned JSON reference data (roles, skills, courses, SDG map, O*NET tasks).

Trend pipeline: arXiv (OpenAlex fallback), Hacker News “Who is hiring?” via the Algolia API, Malaysian job ads via freehire.me, OpenDOSM (employment by occupation, labour force, sector, prices), and Stack Overflow, GitHub, npm and PyPI developer activity. Three years of history per series become damped-trend forecasts to 2030 with 80% ranges and a published backtest; each snapshot waits for an admin's approval.

Hosting and operations: Vercel (frontend), Render (API, Docker, Singapore), GitHub Actions (tests on SQLite and Postgres plus the build on every push; scheduled data refresh), Sentry error tracking.

System block diagram: Figure 1 below.

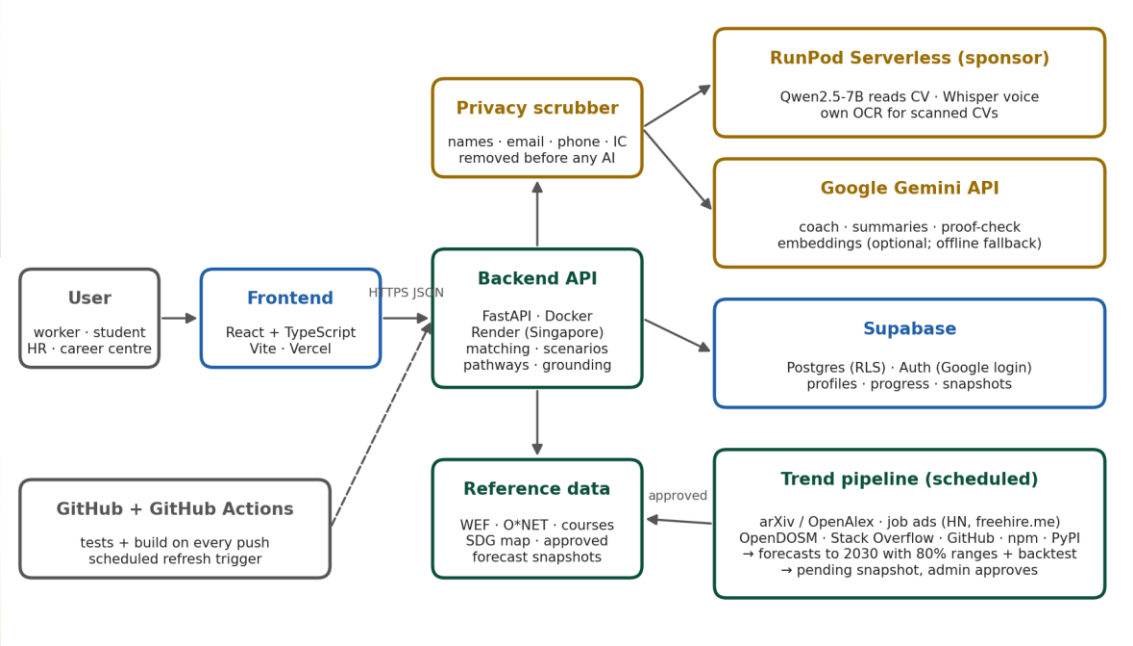


Figure 1. Laluan system architecture: the workshop elements (GenAI APIs, database, cloud deployment) in one deployed system.

Sponsor resource: Yes. RunPod (SDC sponsor). Our RunPod Serverless GPU endpoints, paid from the sponsor credit, run Qwen2.5-7B-Instruct (CV reading), faster-whisper (voice input) and our OCR worker (scanned CVs). Running these on open models we control keeps raw CV text and audio away from third-party AI APIs. Gemini is called through the team's own Gemini API key.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise. Personas from the brief's users: Aisyah, a clerk facing an AI invoice reader; Wei Jie, a final-year student; an HR team planning reskilling. Their fears set the tone: calm, honest, never alarmist.

Define. A ranked, explainable plan in under five minutes, every number traceable to a dated source.

Ideate. We rejected a pure chatbot because its answers cannot be checked. AI reads CVs and explains; forecasts come from public data with a published method.

Prototype and test. Live since Phase 1. Browser testing fed back into the design: a phone layout, a “server waking up” notice, human approval of trend data. A backtest exposed that plain trend lines overstate change, so we added shrinkage and damping. Every fix is in REVIEW_LOG.md.

Deploy. Vercel (frontend), Docker on Render (API), Supabase (data), tests on every push. It works without an API key. Phase 2: human evaluation with a counsellor and native speakers via the in-app review kit.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**4.4** "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." **8.5** "By 2030, achieve full and productive employment and decent work for all women and men, including for young people…" Supporting: 4.3 (affordable technical and vocational education) and 8.2 (productivity through innovation).
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Only **26%** of Malaysian professionals using AI at work have had formal AI training (Hays Asia Salary Guide 2025, as cited in the Domain 1 brief). **600,000+** Malaysian workers need reskilling within 3–5 years (MOSTI, same source). Data-entry clerk jobs are projected to fall **≈26%** by 2030 (WEF Future of Jobs Report 2025, Fig. 2.3). For an individual today, no source-cited, personalised pathway exists in one place: the share of recommendations a user can trace to a source is effectively **0%** in generic chatbot advice.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	(1) Time from CV to a ranked, costed pathway (minutes). (2) Skill-extraction recall on 5 synthetic CVs marked by a person (%). (3) Share of role, skill, task and course recommendations that show a dated source (%). (4) Share of plan steps that are free or free to audit (%).

Metric Category	Your Response
	(5) Reviewer agreement with the top-3 roles for the two personas (%). (6) Forecast backtest: share of clear skill trends whose direction held in the held-out year (%).
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	(1) ≤ **5 minutes** in a timed run. (2) ≥ **80%** average recall. (3) **100%** of recommendations cited (already enforced by automated tests). (4) ≥ **70%** of steps free or free to audit in both demo pathways (currently 2 of 2 for Aisyah and 1 of 1 for Wei Jie). (5) ≥ **70%** reviewer agreement from a lecturer or career counsellor. (6) Better than the "everything rises" guess, re-measured at every refresh (21 Oct 2026: 14 of 19 clear trends held, against 12 for "everything rises").

Sources: World Economic Forum, Future of Jobs Report 2025, January 2025, Figures 2.1 and 2.3, https://reports.weforum.org/docs/WEF_Future_of_Jobs_Report_2025.pdf. SDC Hackathon, Domain 1 Problem Statement & Marking Rubric (citing Hays Asia Salary Guide 2025 and MOSTI). United Nations, SDG targets 4.3, 4.4, 8.2 and 8.5, https://sdgs.un.org/goals.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Not applicable: all code, design, data and documents were created by the team during the hackathon. The full history is public in the code repository.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React 18.3.1, React DOM 18.3.1, Vite 5.4.11, @vitejs/plugin-react 4.3.4	Code	MIT
TypeScript 5.6.3	Code	Apache-2.0
@supabase/supabase-js 2.45.4, @sentry/react 8.38.0, qrcode 1.5.4	Code	MIT
Inter font (@fontsource-variable/inter 5.3.0)	Code (font)	SIL OFL 1.1
FastAPI, Starlette, Pydantic, Uvicorn, HTTPX	Code	MIT / BSD-3 / MIT / BSD-3 / BSD-3
google-genai SDK, python-multipart	Code	Apache-2.0

pypdf, psycopg 3, PyJWT, sentry-sdk	Code	BSD-3 / LGPL-3.0 / MIT / MIT
RapidOCR, ONNX Runtime, pypdfium2, Pillow (OCR for scanned CVs)	Code	Apache-2.0 / MIT / Apache-2.0 or BSD-3 / MIT-CMU
PyMuPDF (one-off WEF figure extraction script, not in the app)	Code	AGPL-3.0
Google Gemini gemini-3.8-flash and gemini-embedding-001 (called via API)	Model	Gemini API Terms of Service
Qwen2.5-7B-Instruct (Alibaba Qwen team), served with vLLM on RunPod	Model	Apache-2.0
OpenAI Whisper small via faster-whisper, on RunPod	Model	MIT
PaddleOCR PP-OCRv4 English recogniser and RapidOCR text detector	Model	Apache-2.0
WEF Future of Jobs Report 2025, Figure 2.3 (figures cited)	Dataset	WEF terms of use, with citation
O*NET 30.0 task statements (U.S. Department of Labor)	Dataset	CC BY 4.0
OpenDOSM / DOSM: wages_occupation, lfs_month, employment_sector, cpi_headline; Salaries & Wages 2025	Dataset	CC BY 4.0
TalentCorp MyCOL 2024/2025 (critical occupations, cited with links)	Dataset	Cited with link
arXiv API counts; OpenAlex	Dataset	arXiv API terms; CC0
Hacker News “Who is hiring?” mention counts (Algolia HN API)	Dataset	HN API terms (counts only)
Malaysian job postings via the freehire.me API	Dataset	freehire terms; project MIT
Stack Overflow, GitHub, npm and PyPI activity counts	Dataset	Each API's terms (counts only)
Anthropic Economic Index	Dataset	CC BY
OpenAI “GPTs are GPTs” occupation exposure scores	Dataset	MIT
ILO Working Paper 140, Annex Table A1	Dataset	CC BY 4.0

UN SDG names, targets and colours	Dataset	UN SDG guidelines
Three landing-page photos (Unsplash: Gema Saputera, Mimi Thian, Vitaly Gariev)	Images	Unsplash License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

Not applicable: no team member is bound by such an agreement.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Signature: Tan Hong Sheng (Team Leader)

Date: 24 October 2026